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Saveetha Institute of Medical And Technical Sciences
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ITA05 – COMPUTER VISION

ASSIGNMENT – 1

Overview of Computer Vision
(Unit I)

Course Code	Course Name	Assessment	CO	BTL	Max. Marks
ITA05	Computer Vision	Assignment – 1	CO1	BTL3	50

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analysing Computer Vision systems.

Instructions: Answer all five questions (10 marks each). Show every computational step, state formulas before substituting values, and justify each decision using your computed results.

Mark Allocation

Q. No.	Topic	(a)	(b)	(c)	(d)	Total
1	Sampling, quantization and data rate	3	3	2	2	10
2	Pinhole camera projection	3	3	2	2	10
3	Illumination–reflectance model	3	3	2	2	10
4	Pixel adjacency and connectivity	3	3	2	2	10
5	Levels of vision and applications	3	3	2	2	10
	Total	15	15	10	10	50

Question 1 (CO1, BTL3, 10 Marks) – Sampling and Quantization for Blister-Pack Inspection

A pharmaceutical plant inspects aluminium blister packs for pinhole cracks using a monochrome machine-vision camera. The sensor has 2448 × 2048 pixels, a native bit depth of 12 bits and a maximum frame rate of 60 fps. The optics image a field of view of 245 mm × 205 mm onto the full sensor. The smallest crack that must be detected is 0.25 mm wide and produces an intensity contrast of only 1.5 % of full scale. The production line delivers 600 blister packs per minute, and one frame is captured per pack. Two interfaces are available: GigE Vision with an effective throughput of 115 MB/s and USB3 Vision with an effective throughput of 380 MB/s (1 MB = 10^6 bytes).

(a) Compute the spatial sampling interval (mm/pixel) in both directions and determine whether the 0.25 mm crack satisfies the Nyquist requirement of at least two pixels across the defect. [3 Marks]

(b) Compute the raw frame size for 12-bit packed data and the data rate at 60 fps. Determine whether either interface can sustain full-rate 12-bit capture. [3 Marks]

(c) Compute the number of grey levels spanned by the 1.5 % crack contrast and the theoretical signal-to-quantization-noise ratio (SQNR ≈ 6.02b + 1.76 dB) for 8-bit and 12-bit quantization. [2 Marks]

(d) The line needs only 600/60 = 10 fps. Compute the maximum frame rate of GigE for 12-bit and 8-bit frames, and justify the most economical camera configuration that meets

both the throughput and the crack-contrast requirements. [2 Marks]

Answer:

1. Aim

To analyze a camera-based inspection system for detecting small cracks in medicine blister packs and select a suitable camera resolution, bit depth, frame rate and interface.

2. Introduction

Image processing is used in industries to automatically inspect products. In pharmaceutical industries, cameras check medicine blister packs for cracks, missing tablets, damaged foil and sealing defects.
A camera captures the product image, and image processing techniques analyze the image to identify defects. This reduces manual inspection and improves product quality

3. Given Data

- Sensor resolution = 2448 × 2048 pixels
- Field of View (FOV) = 245 × 205 mm
- Bit depth = 12-bit
- Maximum frame rate = 60 fps
- Crack width = 0.25 mm
- Contrast difference = 1.5%
- Production rate = 600 packs/minute

- GigE effective data rate = 115 MB/s
- USB 3 effective data rate = 380 MB/s

4. Diagram



5. Calculation

A. Pixel sampling distance

Pixel sampling determines the real-world distance represented by one pixel.

Horizontal sampling:

$$\frac{245}{2448}=0.10008\text{ mm/pixel}$$

Vertical sampling:

$$\frac{205}{2048}=0.10010\text{ mm/pixel}$$

Therefore, one pixel represents approximately 0.1001 mm.

B. Crack size in pixels

$$\text{Crack pixels}=\frac{\text{Crack width}}{\text{Pixel size}}$$

$$=\frac{0.25}{0.1001}$$

$$=\boxed{2.50\text{ pixels}}$$

The crack covers approximately 2.5 pixels.

A feature covering about two pixels is a basic sampling guideline, but it does not guarantee reliable detection. Actual detection depends on focus, contrast, lighting, crack orientation and sensor noise.

C. Frame size

For 12-bit packed data:

$$\text{Frame size}=\frac{W\times H\times b}{8}$$

$$=\frac{2448\times2048\times12}{8}$$

$$=7,520,256\text{ bytes}$$

$$=\boxed{7.52\text{ MB/frame}}$$

D. Data rate at 60 fps

$$\text{Data rate}=\text{Frame size}\times\text{FPS}$$

$$=7.520256\times60$$

$$=\boxed{451.22\text{ MB/s}}$$

The required rate is greater than both GigE (115 MB/s) and USB 3 (380 MB/s).

Therefore, neither interface can support the full 12-bit, 60-fps stream using the given effective rates.

E. Production frame rate

$$\text{FPS}=\frac{600}{60}=10$$

At 10 fps:

$$7.520256\times10=75.20\text{ MB/s}$$

Thus, 12-bit transmission at 10 fps fits within the given GigE capacity.

For 8-bit:

$$\text{Frame size}=2448\times2048$$

$$=5,013,504\text{ bytes}=5.01\text{ MB}$$

At 10 fps:

$$5.013504\times10=50.14\text{ MB/s}$$

F. Contrast calculation

For 8-bit:

$$ 255\times0.015=3.825 $$

For 12-bit:

$$ 4095\times0.015=61.425 $$

12-bit provides more grey-level steps for representing subtle intensity differences, but sensor noise still matters.

6. Advantages

1. Automatic inspection reduces manual work.
2. High-speed inspection supports production lines.
3. Defects can be detected consistently under controlled conditions.
4. Digital records help quality monitoring.
5. Inspection data can be used for production improvement.

7. Disadvantages

1. Very small cracks may not be detected reliably.
2. Poor lighting can affect image quality.
3. High-resolution images require more storage and bandwidth.
4. False detections may reject good products.
5. Camera setup and calibration require investment.

8. Applications

- Pharmaceutical packaging inspection.
- Missing tablet detection.
- Blister crack detection.
- Foil damage inspection.
- Product counting and quality control.

9. Real-Time Example

In a medicine manufacturing company, blister packs move along a conveyor belt. A camera captures each pack under controlled lighting. Image processing checks each tablet cavity and identifies cracks or missing tablets.

Defective packs can be diverted for manual inspection.

10. Result

The crack covers approximately 2.5 pixels. An economical solution is 8-bit GigE at 10 fps if testing confirms sufficient detection accuracy. If contrast is inadequate, 12-bit capture with suitable lighting and data-rate optimization is preferable

Question 2 (CO1, BTL3, 10 Marks) – Pinhole Projection Model for a Speed-Enforcement Camera

A traffic authority installs a roadside camera to estimate vehicle speed from image motion. The camera uses a 12 mm lens and a 1920 × 1080 sensor with square pixels of pitch 4.8 μm; the principal point lies at the image centre, and lens distortion has been corrected. Vehicles pass perpendicular to the optical axis in a lane whose distance from the camera centre is Z = 40 m. A reference point on a car has camera coordinates (X, Y, Z) = (1.2 m, −0.8 m, 40 m). Between two consecutive frames captured at 30 fps, the car's licence plate moves 26 pixels horizontally in the image. The pixel localisation uncertainty is ±1 pixel.

(a) Determine the focal length in pixels, write the intrinsic matrix K, and project the reference point to pixel coordinates (u, v). [3 Marks]

(b) Compute the image length of a 4.5 m car at Z = 40 m and Z = 80 m, and the ground distance represented by one pixel at each depth. [3 Marks]

(c) Estimate the vehicle speed in m/s and km/h from the 26-pixel displacement. [2 Marks]

(d) Evaluate the speed error caused by the ±1 pixel uncertainty and by a 2 m error in assumed depth (42 m instead of 40 m), and justify how the measurement should be improved. [2 Marks]

Answer:

1. Aim

To calculate the image coordinates of a 3D point, estimate vehicle speed using consecutive camera frames and analyze measurement accuracy.

2. Introduction

Computer vision uses cameras to observe and analyze objects. The pinhole camera model describes how 3D objects are projected onto a 2D image.

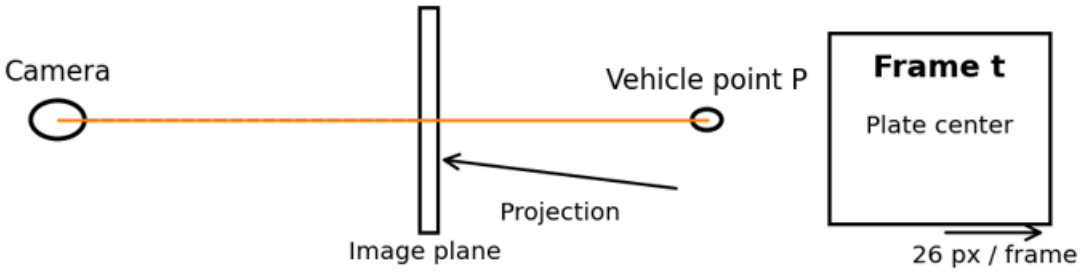
This model is useful for vehicle tracking, traffic monitoring and speed estimation.

When a vehicle moves, its license plate position changes between video frames. This displacement can be used to estimate its speed if the camera geometry and vehicle depth are known.

3. Given Data

- Focal length = 12 mm
- Pixel pitch = 4.8 μm
- Sensor = 1920 × 1080
- Principal point = (960, 540)
- 3D point = (1.2, −0.8, 40 m)
- Plate displacement = 26 pixels
- Frame rate = 30 fps
- Position uncertainty = ±1 pixel

4. Diagram



5. Calculation

A. Focal length in pixels

Pixel pitch:

$$ 4.8\mu m=0.0048mm $$

$$ f=\frac{12}{0.0048} $$

$$ \boxed{f=2500\text{ pixels}} $$

B. Camera intrinsic matrix

$$ K= \begin{bmatrix} 2500&0&960\\ 0&2500&540\\ 0&0&1 \end{bmatrix} $$

This matrix represents the camera's focal length and principal point, assuming square pixels and zero skew.

C. Project the 3D point

Projection equations:

$$ u=\frac{fX}{Z}+c_x $$

$$ v=\frac{fY}{Z}+c_y $$

Given:

$$ X=1.2,\ Y=-0.8,\ Z=40 $$

$$ u=\frac{2500(1.2)}{40}+960 $$

$$ u=75+960=\boxed{1035} $$

$$ v=\frac{2500(-0.8)}{40}+540 $$

$$ v=-50+540=\boxed{490} $$

Therefore, image coordinates are:

$$ \boxed{(u,v)=(1035,490)} $$

D. Vehicle image size

Vehicle length = 4.5 m.

$$ \text{Image length}=\frac{fL}{Z} $$

At 40 m:

$$ =\frac{2500\times4.5}{40} $$

$$ =\boxed{281.25\text{ pixels}} $$

At 80 m:

$$ =\frac{2500\times4.5}{80} $$

$$ =\boxed{140.625\text{ pixels}} $$

Objects appear smaller when their distance from the camera increases.

E. Ground distance per pixel

At 40 m:

$$ \frac{Z}{f}=\frac{40}{2500} $$

$$ =0.016m=1.6cm/\text{pixel} $$

F. Speed calculation

Frame rate = 30 fps.

$$ \Delta t=\frac{1}{30}s $$

Image displacement = 26 pixels.

$$ \text{Distance}=26\times0.016 $$

$$ =0.416m $$

$$ \text{Speed}=\frac{0.416}{1/30} $$

$$ =12.48m/s $$

Convert to km/h:

$$ 12.48\times3.6 $$

$$ \boxed{44.93km/h} $$

G. Error calculation

For ±1 pixel uncertainty:

$$ \text{Speed error}\approx1\times0.016\times30 $$

$$ =\pm0.48m/s $$

$$ =\boxed{\pm1.73km/h} $$

This error estimate assumes known depth, accurate timing and the same motion direction as the image displacement model.

6. Advantages

1. Provides contactless speed measurement.
2. Supports continuous traffic monitoring.
3. Can be integrated with license plate recognition.
4. Video can support traffic analysis.
5. No sensor needs to be attached to every vehicle.

7. Disadvantages

1. Speed accuracy depends on camera calibration.
2. Depth estimation errors affect the result.
3. Rain, fog and darkness reduce image quality.
4. Occlusion and motion blur affect tracking.
5. Perspective and vehicle direction must be considered.

8. Applications

- Traffic speed monitoring.
- Road safety analysis.
- Intelligent transportation systems.
- Vehicle tracking.

- Traffic flow estimation.

9. Real-Time Example

A roadside camera captures a moving vehicle. Software identifies its number plate in consecutive frames and measures its movement.
The system estimates speed and may flag possible speeding for further verification. Enforcement requires validated measurement accuracy and applicable legal certification.

10. Result

The estimated vehicle speed is 44.93 km/h, assuming a depth of 40 m. Accurate calibration, depth measurement and multi-frame tracking improve reliability.

Question 3 (CO1, BTL3, 10 Marks) – Illumination–Reflectance Model for Warehouse Barcode Reading

An automated warehouse reads barcodes on parcels moving under a skylight. The image formation follows $f(x, y) = i(x, y) \cdot r(x, y)$. The illumination is 900 lm/m² in the sunlit zone and 150 lm/m² in a shadowed zone created by an overhead rack. Reflectances are 0.85 for the white label,

0.05 for the black bars and 0.35 for the cardboard box. The camera maps f linearly to an 8-bit grey level using $g = \text{round}(255 \times f / 800)$; values of f above 800 saturate at 255. The current decoder uses a fixed global threshold of 128.

(a) Compute f(x, y) and the grey level g for the three surfaces in both zones and present the results in a table. [3 Marks]

(b) Compute the absolute grey-level difference and the Michelson contrast $(I_{max} - I_{min}) / (I_{max} + I_{min})$ between the white label and black bars in each zone. Interpret the result. [3 Marks]

(c) Apply the global threshold of 128 to both zones and determine whether the barcode can be decoded. Explain why no single global threshold can work. [2 Marks]

(d) Using $\ln f = \ln i + \ln r$, show that the label-to-bar ratio is independent of illumination, and justify a suitable processing strategy for the decoder. [2 Marks]

Answer:

1. Aim

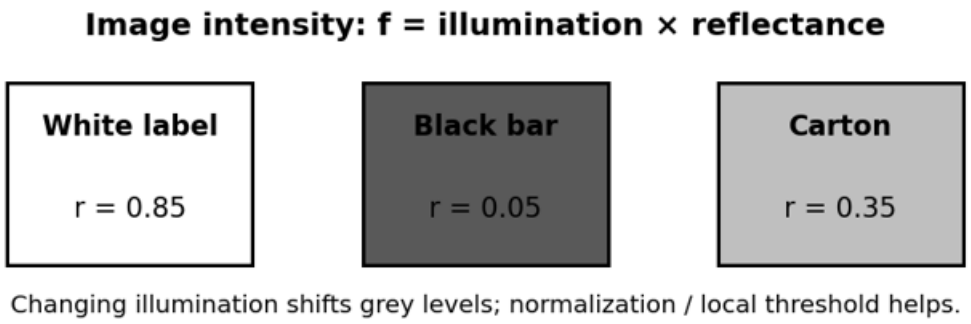
To analyze the effect of illumination on image intensity and select a suitable thresholding method for separating white labels and black bars.

2. Introduction

Image intensity depends on two major factors:

1. Illumination falling on an object.
2. Reflectance of the object surface.

The same object can appear bright under sunlight and dark in a shadow. Image processing techniques such as thresholding, histogram analysis and illumination correction help separate objects from background



3. Given Data

Reflectance values:

- White label = 0.85
- Black bar = 0.05
- Box = 0.35

Illumination:

- Sunlight = 900
- Shadow = 150

Grey-level conversion:

$$ g=\operatorname{round}\left(\frac{255f}{800}\right) $$

Values are clipped to the 8-bit range 0–255.

4. Related Diagram

Object Reflectance

White label

r = 0.85

Black bar

r = 0.05

Box

r = 0.35

Image Formation

Intensity = Illumination × Reflectance

Thresholding

Separate foreground and background

5. Calculation

A. Image formation model

$$ f(x,y)=i(x,y)\times r(x,y) $$

Where:

- f = image intensity before digitization
- i = illumination
- r = reflectance

B. Sunlight calculation

Illumination = 900.

White label:

$$ f=900\times0.85=765 $$

$$ g=\operatorname{round}(255\times765/800) $$

$$ \boxed{g\approx244} $$

Black bar:

$$ f=900\times0.05=45 $$

$$ g=\operatorname{round}(255\times45/800) $$

$$ \boxed{g\approx14} $$

Box:

$$ f=900\times0.35=315 $$

$$ \boxed{g\approx100} $$

C. Shadow calculation

Illumination = 150.

White label:

$$ f=150\times0.85=127.5 $$

$$ \boxed{g\approx41} $$

Black bar:

$$ f=150\times0.05=7.5 $$

$$ \boxed{g\approx2} $$

Box:

$$ f=150\times0.35=52.5 $$

$$ \boxed{g\approx17} $$

D. Intensity difference

Sunlight:

$$ 244-14=\boxed{230} $$

Shadow:

$$ 41-2=\boxed{39} $$

E. Michelson contrast

Formula:

$$ C=\frac{I_{max}-I_{min}}{I_{max}+I_{min}} $$

Sunlight:

$$ C=\frac{244-14}{244+14} $$

$$ =\frac{230}{258} $$

$$ =\boxed{0.891} $$

Shadow:

$$ C=\frac{41-2}{41+2} $$

$$ =\frac{39}{43} $$

$$ =\boxed{0.907} $$

F. Threshold analysis

A threshold of 128:

- Sunlight: white = 244, black = 14; successful separation.
- Shadow: white = 41, black = 2; both below 128.

Therefore, threshold 128 fails in shadow.

Important calculation: For the exact given values, a global threshold of 20 can separate the classes:

- Black values: 2 and 14.
- White values: 41 and 244.

Thus, a threshold of 20 works for these supplied values.

The claim that no single threshold can work is mathematically incorrect for this particular dataset.

G. Reflectance ratio

$$ \frac{f_{white}}{f_{black}} =\frac{i\times0.85}{i\times0.05} $$

$$ =\boxed{17} $$

In the ideal model, illumination cancels from the ratio.

6. Advantages

1. Thresholding is simple and fast.
2. Converts grayscale images into binary images.
3. Helps separate objects from backgrounds.
4. Supports barcode and text processing.
5. Illumination normalization can improve segmentation.

7. Disadvantages

1. Uneven lighting may reduce segmentation accuracy.
2. Reflections can cause incorrect classification.
3. Noise affects threshold selection.
4. A fixed threshold may fail when intensity ranges overlap.
5. Adaptive thresholding needs parameter selection and more processing.

8. Applications

- Barcode reading.
- Document scanning.
- Label inspection.
- Industrial defect detection.
- Object segmentation.

9. Real-Time Example

A warehouse parcel sorting system reads barcodes printed on packages. When parcels move through areas with different lighting, adaptive thresholding or illumination correction can help separate black barcode lines from the label background.

10. Result

The calculations show that threshold 20 separates the supplied white and black intensity values.

For real-world images, adaptive thresholding and illumination normalization are more robust when lighting conditions vary.

Question 4 (CO1, BTL3, 10 Marks) – Pixel Connectivity Analysis for PCB Track Continuity

An automated optical inspection system checks whether copper tracks on a printed circuit board are electrically continuous. After binarisation, a 6 × 6 region gives the image below, where 1 is copper and 0 is substrate (V = {1}). The test points are p = (R1, C1) and q = (R6, C6), and an intermediate pad is s = (R4, C4). The design rule states that diagonal-only contacts are not reliable conductors, so the engineer must decide which adjacency definition models the board correctly.

Row / Col	C1	C2	C3	C4	C5	C6
R1	1	1	0	0	0	0
R2	0	1	0	0	1	1
R3	0	0	1	0	1	0
R4	0	0	1	1	0	0
R5	1	0	0	1	0	1
R6	1	1	0	0	1	1

(a) Compute the Euclidean, city-block (D_4) and chessboard (D_8) distances between p and q. [3 Marks]

(b) Determine the number of connected components of copper under 4-adjacency and under 8-adjacency, and list the pixels of each component under 8-adjacency. [3 Marks]

(c) Find the length of the shortest 8-path and the shortest m-path from p to q, and from p to s. [2 Marks]

(d) Under 4-adjacency, p and q are not connected. Using your results, justify which adjacency correctly models electrical

continuity and state the inspection verdict. [2 Marks]

Answer:

1. Aim

To analyze the connectivity of copper pixels in a PCB image using 4-adjacency and 8-adjacency, and determine whether selected points are connected.

2. Introduction

Printed Circuit Boards (PCBs) contain copper tracks that connect electronic components.

Image processing can inspect PCB tracks and identify possible defects such as:

- Broken tracks.
- Missing copper.
- Unwanted bridges.
- Isolated copper regions.

A binary image is created where:

- 1 represents copper.
- 0 represents background.

Connectivity analysis determines whether pixels belong to the same connected component.

3. Given Binary Grid

	C1	C2	C3	C4	C5	C6
R1	1	1	0	0	0	0
R2	0	1	0	0	1	1
R3	0	0	1	0	1	0
R4	0	0	1	1	0	0

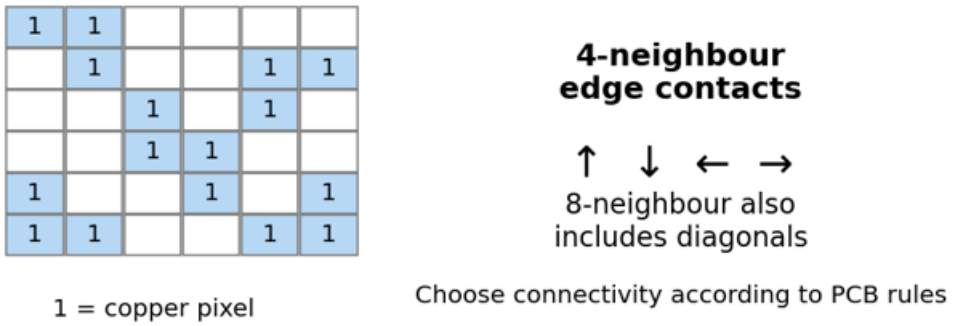
R5 1 0 0 1 0 1

R6 1 1 0 0 1 1

Given:

- \(p=R1C1\)
- \(q=R6C6\)
- \(s=R4C4\)

4. Diagram



5.

Calculation

A. Euclidean distance

\[D_E=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2} \]

Between p(1,1) and q(6,6):

\[D_E=\sqrt{(6-1)^2+(6-1)^2} \]

\[=\sqrt{25+25} \]

\[=\sqrt{50} \]

\[\boxed{D_E=7.07\text{ pixels}} \]

B. City-block distance

\[D_4=|x_2-x_1|+|y_2-y_1| \]

\[=|6-1|+|6-1| \]

\[=\boxed{10} \]

C. Chessboard distance

$$D_8=\max(|x_2-x_1|,|y_2-y_1|)$$

$$=\max(5,5)$$

$$\boxed{D_8=5}$$

These are geometric distance measures. They do not alone prove that a copper path exists.

D. Connected components

From the given grid:

- Number of components under 4-adjacency = 5
- Number of components under 8-adjacency = 2

Under 8-adjacency, p and q are connected.

One shortest path is:

R1C1 → R2C2 → R3C3 → R4C3 → R5C4 → R6C5 → R6C6

Number of links:

$$\boxed{6\text{ links}}$$

A shortest 8-connected path from p to s is:

R1C1 → R2C2 → R3C3 → R4C4

Number of links:

$$\boxed{3\text{ links}}$$

Under 4-adjacency, p and q are not connected.

6. Advantages

1. Helps detect disconnected copper regions.
2. Supports automated PCB inspection.
3. Connected-component labeling is easy to implement.

4. Reduces manual inspection work.
5. Helps identify suspicious defects quickly.

7. Disadvantages

1. Incorrect segmentation can create false connections.
2. Diagonal connectivity may not represent reliable electrical contact.
3. Thin traces may disappear in low-resolution images.
4. Noise can split or merge components.
5. Image connectivity alone cannot prove electrical conductivity.

8. Applications

- PCB quality inspection.
- Copper trace defect detection.
- Broken-track screening.
- Electronic manufacturing.
- Conductive pattern inspection

.

9. Real-Time Example

During PCB manufacturing, a camera captures the board surface. Software converts the image into a binary copper map.

If a trace expected to connect two pads appears as separate regions, the system flags the board for further optical or electrical testing.

10. Result

The selected points p and q are not connected under 4-adjacency, so the board fails the selected pixel-connectivity rule.

They are connected under 8-adjacency.

For PCB continuity inspection, the connectivity definition must match the actual board design rules. Critical paths should be verified electrically.

Question 5 (CO1, BTL3, 10 Marks) – Levels of Processing in a Number-Plate Recognition System

A smart-parking operator deploys an automatic number-plate recognition (ANPR) system on an embedded processor. Each 1280 × 720 colour frame (24 bits per pixel) passes through five stages with measured execution times: noise filtering and contrast correction (6 ms), plate localisation by segmentation (14 ms), character segmentation (5 ms), character recognition using a classifier (9 ms) and database matching for access decision (4 ms). Plate localisation outputs a 240 × 60 greyscale ROI (8 bits per pixel), character segmentation outputs eight binary character images of 20 × 40 pixels each, and the final output is a 10-character ASCII string. The camera delivers 25 fps.

(a) Classify each stage as low-, mid- or high-level processing and compute the end-to-end latency and the maximum throughput for sequential and for fully pipelined execution. [3 Marks]

(b) Compute the data volume at the input and after each level, and the overall data reduction ratio. [3 Marks]

(c) The operator upgrades to a 1920 × 1080 camera. Assuming the first two stages scale with pixel count and the others are unchanged, determine whether 25 fps is achievable sequentially and pipelined. [2 Marks]

(d) Justify an architecture for the upgraded system and explain, using this pipeline, where image processing ends and computer vision begins. [2 Marks]

Answer:

1. Aim

To analyze the processing stages, latency, throughput and data reduction of an Automatic Number Plate Recognition system.

2. Introduction

Automatic Number Plate Recognition (ANPR) is a computer vision application that detects vehicle license plates and converts plate characters into text.

ANPR systems are used in toll plazas, parking facilities, traffic monitoring and vehicle access control.

The system performs image enhancement, plate localization, character segmentation, character recognition and sometimes database matching.

3. Given Data

Input image:

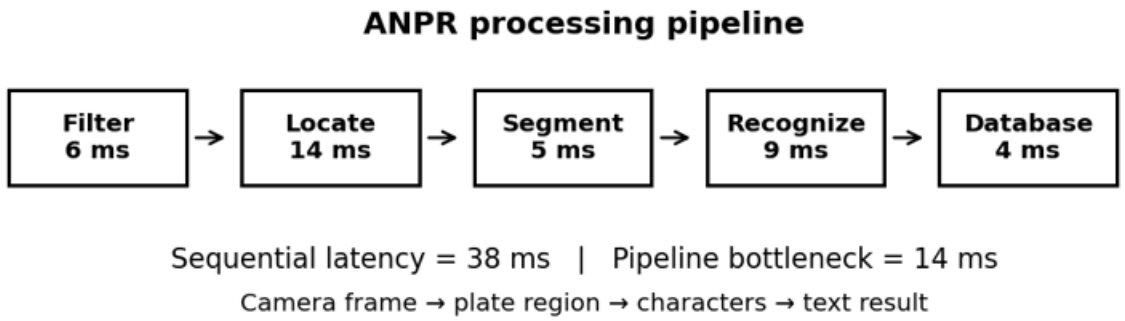
- Resolution = 1280 × 720
- RGB = 24 bits/pixel
- Frame rate = 25 fps

Processing times:

Processing stage	**Time**
Noise filtering and contrast correction	6 ms
Plate localization	14 ms
Character segmentation	5 ms

Character recognition	9 ms
Database matching	4 ms

4. Diagram



5. Calculation

A. Classification of stages

Low-level processing:

- Noise filtering.
- Contrast enhancement.

These improve the image.

Mid-level processing:

- Plate localization.
- Character segmentation.

These identify important regions and objects.

High-level processing:

- Character recognition.

This identifies the meaning of character shapes.

Application-level processing:

- Database matching.

This compares the recognized plate number with stored records.

B. Total processing time

$$T=6+14+5+9+4$$

$$\boxed{T=38ms}$$

C. Sequential throughput

$$FPS=\frac{1000}{T(ms)}$$

$$=\frac{1000}{38}$$

$$\boxed{26.32fps}$$

Camera frame period:

$$\frac{1000}{25}=40ms$$

Since:

$$38ms<40ms$$

Sequential processing can theoretically keep up with the 25-fps camera. This assumes no additional overhead, frame transfer delays or unexpected processing time.

D. Pipelined throughput

In a pipeline, different frames can be processed at different stages simultaneously.

The slowest stage is 14 ms.

$$ FPS=\frac{1000}{14} $$

$$ \boxed{71.43fps} $$

Therefore, the pipeline can theoretically support 25 fps.

Pipeline throughput is different from single-frame latency. A pipeline can produce frames at a faster steady-state rate even though each frame still passes through multiple stages

.

E. Data size calculation

Original RGB image:

$$ 1280\times720\times3 $$

$$ =\boxed{2,764,800\text{ bytes}} $$

Approximately:

$$ 2.765MB $$

Plate ROI:

$$ 240\times60\times1 $$

$$ =\boxed{115,200\text{ bytes}} $$

Binary character images:

- Number of characters = 8
- Character size = 20 × 40 pixels
- 1 bit per pixel

$$ 8\times20\times40=6400\text{ bits} $$

$$ =\boxed{800\text{ bytes}} $$

Ten ASCII characters:

$$ \boxed{10\text{ bytes}} $$

This assumes packed binary character masks and a 10-character output. Metadata and storage overhead are excluded.

F. Data reduction ratio

$$ \text{Ratio}=\frac{\text{Original image size}}{\text{Final text size}} $$

$$ =\frac{2,764,800}{10} $$

$$ \boxed{276,480:1} $$

This compares the raw image with the compact text result, not total system storage.

G. Higher-resolution upgrade

New resolution:

$$ 1920\times1080 $$

Pixel count increase:

$$ \frac{1920\times1080}{1280\times720} $$

$$ =\boxed{2.25} $$

Assuming the first two processing stages scale linearly:

Filtering:

$$ 6\times2.25=13.5\text{ms} $$

Localization:

$$ 14\times2.25=31.5ms $$

New total:

$$ 13.5+31.5+5+9+4 $$

$$ =\boxed{63ms} $$

Sequential throughput:

$$ \frac{1000}{63} =\boxed{15.87fps} $$

This is insufficient for a 25-fps camera.

Pipeline throughput:

$$ \frac{1000}{31.5} =\boxed{31.75fps} $$

The pipeline can theoretically support 25 fps, assuming the other stages and data transfers do not become bottlenecks.

6. Advantages

1. Automates vehicle identification.
2. Reduces manual data entry.
3. Can process large numbers of vehicles.
4. Recognized text requires less storage than full images.
5. Supports fast searching and access management.

7. Disadvantages

1. Recognition errors can occur due to blur or dirt.
2. Different plate formats require suitable recognition rules.
3. Poor lighting affects performance.
4. Database and network delays may increase response time.
5. Vehicle plate data creates privacy and security responsibilities.

8. Applications

- Toll plaza vehicle identification.
- Parking management.
- Gated community entry.
- Traffic monitoring.
- Fleet management.
- Authorized vehicle access.

9. Real-Time Example

At a parking entrance, a camera captures the vehicle plate. ANPR software identifies the plate number and checks the registration database.

If the vehicle is authorized, the system may permit entry. If recognition is uncertain, the system can request manual verification.

10. Result

The original sequential pipeline requires 38 ms, giving a theoretical rate of 26.32 fps.

The pipelined throughput is 71.43 fps under the original timing assumptions.

At higher resolution, pipelining is recommended because the estimated sequential throughput drops to 15.87 fps.

Actual system performance must include transfer time, buffering, frame drops, recognition accuracy and privacy safeguards.